

**Luiss**

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# Expressions and Data Types

## Introduction to Computer Programming

**Management and Artificial Intelligence**

LUISS



# Why Python?

- A **high-level** language for **beginners**
- with a **clean syntax**, but...
- **Industrial strength** programming language used at thousands of companies (one of three official Google languages)
- Free, well documented, well supported

**NOTE:** We will use Python 3. Python 2 is now unsupported, and there are non-backwards compatible differences between Python 3 and Python 2.

# Expressions and Statements

A (Python) program is composed by a sequence of:

- **definitions** (also **expressions**) which are **evaluated**
- **statements** (also **commands**) which are **executed**

In plain terms...

- **expressions** yield some results (e.g.,  $5 * 5$ )
- **statements** instruct the interpreter to do something (e.g., print on screen, assign to variable)

# Expressions vs. Statements

- **Expressions:** **Represent** something
  - Python *evaluates* expressions
  - end-result is a value

```
>>> 2.3          # a plain literal
2.3
>>> (7*3+1)-4    # 4 literals and some operators
18
```

- **Statements:** **Do** something
  - Python *executes* statements
  - no need to return a value

```
>>> print("Hi there!")
Hi there!
>>> import sys
```

# Interactive vs. Script Mode vs. Jupyter Notebook

- **Interactive Mode:** perfectly fine for testing and learning.

Type your commands/expressions into the shell and Python will execute/evaluate them:

|           |         |
|-----------|---------|
| >>> 5 * 5 | ←input  |
| 25        | ←output |

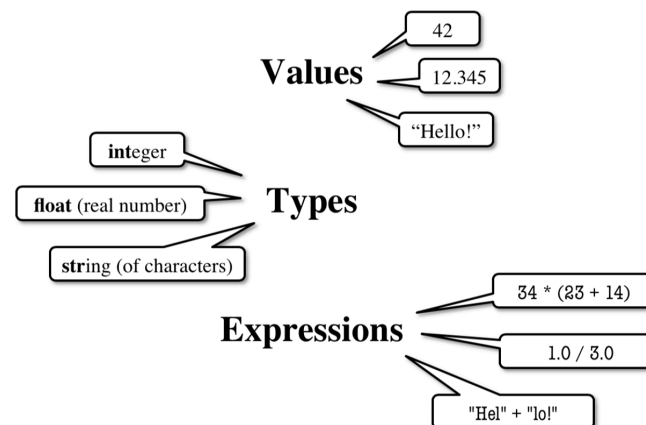
- **Script Mode:** is what you normally do in the real world.  
Write the entire source code in a text file ending with the .py extension (say `myscript.py`).  
`myscript.py` will be executed in its entirety by the Python interpreter
- **Jupyter Notebook:** what we will in this course most of the time.  
Write the code into a code cell and execute it

```
print(5 * 5)
```

25

# Programming in Python

- programs manipulate **data objects**
- each object has its own **type**, which defines the type of operations that can be done with them
- data objects can either be:
  - **Scalar**: primitive data types that cannot be subdivided (like numbers, booleans)
  - **Non-scalar**: composite data types with internal structure that can be accessed (like lists, dictionaries, objects)



# Scalar Objects

- `int`: represents integers (e.g., 5)
- `float`: represents real numbers (e.g., 3.14)
- `bool`: represents Boolean values (True or False)
- `NoneType`: special with only one value (None)

**HINT:** Use the `type()` function if you are not sure about the data type.

Examples:

```
type(109)
```

`int`

```
type(10.9)
```

`float`

# Type `int()`

Values:  $\dots, -3, -2, -1, 0, 1, 2, 3, \dots$

Integer literals look like a sequence of numbers with no periods nor commas.

Operations:

- `+`, `-`, `*`, `/`, `//` (floor division), `%` (modulus, remainder), `**` (exponentiation)
- `-` (unary minus, to negate a number)

Examples:

```
7/5
```

```
1.4
```

```
7//5 # NOTE: // is the floor division!
```

```
1
```



# Type float ( )

Values: some (approximated) range of  $\mathbb{R}$

Note that:

- numbers with a "." are decimals (e.g., 2.0)
- numbers with no decimal separator are integers (e.g., 2)

Operations:

- +, -, \*, /, // (floor division), % (modulus, remainder of division), \*\* (exponentiation)
- - (unary, invert sign)

**NOTE:** Operations between floats or between a float and an integer, always yield a float.

```
7.0/5.0
```

1.4

```
7.0//5
```

1.0

Python stores **floating point numbers** as base 2 (binary) fractions:

$$\{\text{number}\} = \{\text{integer mantissa}\} \cdot 2^{-\{\text{exponent}\}}$$

For example:  $0.125 = 1 \cdot 2^{-3} = 1 \cdot \frac{1}{2^3} = 1 \cdot \frac{1}{8}$ .

**WARNING:** Not all numbers can be represented in base 2 fractions. Python chooses the best approximation it can. The same also holds for some numbers in base 10 fraction: e.g., what about 1/3?

Python may yield some **approximation errors**, that propagate as expressions go through.

Example:

```
0.1+0.2
```

```
0.30000000000000004
```

```
0.1+0.2+0.1+0.2
```

```
0.6000000000000001
```

```
0.1+0.2+0.1+0.2+0.1+0.2+0.1+0.2
```

```
1.2000000000000002
```

# Type bool

Values:

True and False (**capitalized!**)

Operations:

$$\text{not } a = \begin{cases} \text{True} & \text{if } a \text{ is False} \\ \text{False} & \text{if } a \text{ is True} \end{cases}$$

$$a \text{ and } b = \begin{cases} \text{True} & \text{if both } b \text{ and } a \text{ are True} \\ \text{False} & \text{otherwise} \end{cases}$$

$$a \text{ or } b = \begin{cases} \text{True} & \text{if } a \text{ is True or } b \text{ is True} \\ \text{False} & \text{otherwise} \end{cases}$$

Booleans are the output of comparisons between items (e.g., int or float):

- Order:  $i < j$ ,  $i > j$ ,  $i \leq j$ ,  $i \geq j$
- (In)equality:  $i \neq j$ ,  $i == j$

## Truth Table between two Boolean variables

| <b>a</b> | <b>b</b> | <b>not a</b> | <b>not b</b> | <b>a and b</b> | <b>a or b</b> |
|----------|----------|--------------|--------------|----------------|---------------|
| True     | True     | False        | False        | True           | True          |
| False    | False    | True         | True         | False          | False         |
| True     | False    | False        | True         | False          | True          |
| False    | True     | True         | False        | False          | True          |

# Type Conversions: explicit

Switching between float and int can be performed via **casting** (**explicit** conversion):

- `float(2)` converts value 2 (integer) to 2.0 (floating point)
- `int(2.9)` converts value 2.9 (floating point) to 2 (integer)

**WARNING** `int()` does not round to the nearest integer: rather, it truncates the decimal part. To round to the nearest integer, use `round()` instead.

# Type Conversions: implicit

Python supports **widening** automatically (**implicit** conversion):

$$\underbrace{\text{bool} \rightarrow \text{int} \rightarrow \text{float}}_{\text{narrow to wide}}$$

- **Widening:** Python does it automatically, if needed. Example:

`1/0.5=2.0` because Python casts `1` to `float`

- **Narrowing:** Python never does it automatically. Because narrowing leads to information loss!

# Expressions

- **Expressions** are combinations of **objects** and **operators**
- an expression yields a **value**, which has its own **type**
- the simplest expression has the form:

<object, operator, object>

## Arithmetic Operations on `int` and `float`: a Recap

| Syntax             | Operator           | Output Type                                 |
|--------------------|--------------------|---------------------------------------------|
| <code>i+j</code>   | Sum                | if either of them is float, result is float |
| <code>i - j</code> | Difference         | if either of them is float, result is float |
| <code>i*j</code>   | Product            | if either of them is float, result is float |
| <code>i/j</code>   | Division           | result is float (always)                    |
| <code>i%j</code>   | Remainder (Modulo) | if either of them is float, result is float |
| <code>i**j</code>  | Power of           | if either of them is float, result is float |
| <code>i//j</code>  | Floor division     | if either of them is float, result is float |



# Comparison Between `ints` and `floats`: a Recap

**NOTE:** Comparisons below evaluate to a `bool`.

| Expression             | Evaluation                                                          |
|------------------------|---------------------------------------------------------------------|
| <code>i &lt; j</code>  | → True if <code>i</code> is smaller than <code>j</code>             |
| <code>i &lt;= j</code> | → True if <code>i</code> is smaller than or equal to <code>j</code> |
| <code>i &gt; j</code>  | → True if <code>i</code> is greater than <code>j</code>             |
| <code>i &gt;= j</code> | → True if <code>i</code> is greater than or equal to <code>j</code> |
| <code>i == j</code>    | → True if <code>i</code> is equal to <code>j</code>                 |
| <code>i != j</code>    | → True if <code>i</code> is not equal to <code>j</code>             |

# Chained Comparisons

Comparisons can be chained arbitrarily and comparisons have the same priority:

$i < j < k$  is the same as  $i < j$  and  $j < k$

Since comparisons do not act in a pairwise fashion, *weird* comparisons like

$i < j > k$

are perfectly valid since it is equivalent to  $(i < j)$  and  $(j > k)$ .

# Operator Precedence

Precedence between operators can be *forced* via parenthesis.

In principle, operators follow the **PEMDAS** order:

1. **P**arenthesis
2. **E**xponential
3. **M**ultiplication and **D**ivision
4. **A**ddition and **S**ubtraction

# The Big Picture of Operator Precedence

1. Parenthesis: ( . . . )
2. Exponentiation: \*\*
3. Unary operators: + and -
4. Binary arithmetic: \*, /, // and %
5. Binary arithmetic: + and -
6. Comparisons: >, <, >=, <=, ==, !=
7. Logical not
8. Logical and
9. Logical or

**NOTE:** Same line means same precedence. Read ties left to right.

**HINT:** In case of emergency, use parenthesis!

